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Reference-based analysis of lung single-cell sequencing reveals a transitional profibrotic macrophage

Dvir Aran[1,7], Agnieszka P. Looney[2,7], Leqian Liu[3,7], Esther Wu[2], Valerie Fong[2], Austin Hsu[4], Suzanna Chak[2], Ram P. Naikawadi[2], Paul J. Wolters[2], Adam R. Abate[3,5,6], Atul J. Butte[1] and Mallar Bhattacharya[2]*

Tissue fibrosis is a major cause of mortality that results from the deposition of matrix proteins by an activated mesenchyme. Macrophages accumulate in fibrosis, but the role of specific subgroups in supporting fibrogenesis has not been investigated in vivo. Here, we used single-cell RNA sequencing (scRNA-seq) to characterize the heterogeneity of macrophages in bleomycin-induced lung fibrosis in mice. A novel computational framework for the annotation of scRNA-seq by reference to bulk transcriptomes (SingleR) enabled the subclustering of macrophages and revealed a disease-associated subgroup with a transitional gene expression profile intermediate between monocyte-derived and alveolar macrophages. These CX3CR1^{+}SiglecF^{+} transitional macrophages localized to the fibrotic niche and had a profibrotic effect in vivo. Human orthologs of genes expressed by the transitional macrophages were upregulated in samples from patients with idiopathic pulmonary fibrosis. Thus, we have identified a pathological subgroup of transitional macrophages that are required for the fibrotic response to injury.

Fibrosing diseases are a major cause of mortality. Whether toxin-related, genetic, malignant, infectious, autoimmune, or idiopathic, they can affect any organ[1]. Lung fibrosis is a particularly vexing clinical problem because of the lack of effective therapies and a poor understanding of its etiology. The most prevalent form, idiopathic pulmonary fibrosis, has a median survival of only 3 years, and approved treatments are limited[2]. With respect to the pathogenesis of fibrosis in multiple organs, attention has focused on fibroblast activation, given the central role of fibroblasts in the deposition of matrix proteins such as collagen[3–6]. However, the mechanisms of fibroblast activation and proliferation are not fully understood.

The healthy adult lung contains two types of resident macrophages: a self-renewing population of embryonically derived macrophages known as alveolar macrophages and a separate population of interstitial macrophages present near the larger airways and in the lung interstitium[7,8]. In injury models of fibrosis, monocyte-derived cells are recruited to tissue sites and have a profibrotic effect[9–11]. However, the diversity of these cells and the mechanisms by which they mediate the profibrotic effect are incompletely understood, in part because the study of specific transcriptomic subsets has been limited by bulk sequencing approaches. In particular, the kinetically evolving heterogeneity inherent to disease cannot be parsed with the limited range of flow cytometric markers used to study populations in bulk. Because scRNA-seq can resolve diversity within cellular populations, we reasoned that it should enhance understanding of macrophage heterogeneity and specify subsets for functional analysis in fibrosis.

A current limitation in single-cell analysis is that there are no prevalent methods for identification of cell types other than manual annotation, which suffers from subjectivity. Moreover, referencing conventional markers is challenged by incomplete messenger RNA detection. We developed a computational tool for unbiased annotation as well as clustering of scRNA-seq, called SingleR (single-cell recognition). SingleR assigns cellular identity for single cell transcriptomes by comparison to reference data sets of pure cell types sequenced by microarray or RNA-sequencing (RNA-seq). Here, we used SingleR to identify macrophages from a mixed sample of lung cells at baseline and after bleomycin-induced lung injury in mice. In addition, we applied a novel hierarchical clustering approach based on SingleR correlations to subcluster macrophages in fibrosis, which enabled the identification of cells with transitional gene expression between monocyte-derived and alveolar macrophages. We used markers identified by the subclustering analysis to investigate the functional properties of these transitional macrophages.

Our studies revealed that monocyte-derived, disease-associated macrophages transitioning to alveolar identity localized to the fibrotic niche and exerted a profibrotic effect. Specific ablation of these macrophages prevented lung fibrosis. Mechanistically, the scar-associated macrophages were the predominant source of the mitogen Pdgf-aa, which was necessary for proliferation of fibroblasts. These studies uncover a role for macrophages in the paracrine regulation of fibroblasts and provide a perspective on myeloid-mesenchymal crosstalk in response to injury.

Results

Reference data sets enhance annotation of scRNA-seq. To improve annotation of cell types in scRNA-seq, we developed a computational method called SingleR, which correlated single-cell transcriptomes

[1]Bakar Computational Health Sciences Institute, University of California, San Francisco, San Francisco, CA, USA. [2]Division of Pulmonary, Critical Care, Allergy, and Sleep Medicine, Department of Medicine, University of California, San Francisco, San Francisco, CA, USA. [3]Department of Bioengineering and Therapeutic Sciences, University of California, San Francisco, San Francisco, CA, USA. [4]Gladstone Institute of Cardiovascular Disease, San Francisco, CA, USA. [5]California Institute for Quantitative Biosciences, University of California, San Francisco, San Francisco, CA, USA. [6]Chan Zuckerberg Biohub, San Francisco, CA, USA. [7]These authors contributed equally: D. Aran, A.P. Looney, L. Liu. *e-mail: mallar.bhattacharya@ucsf.edu

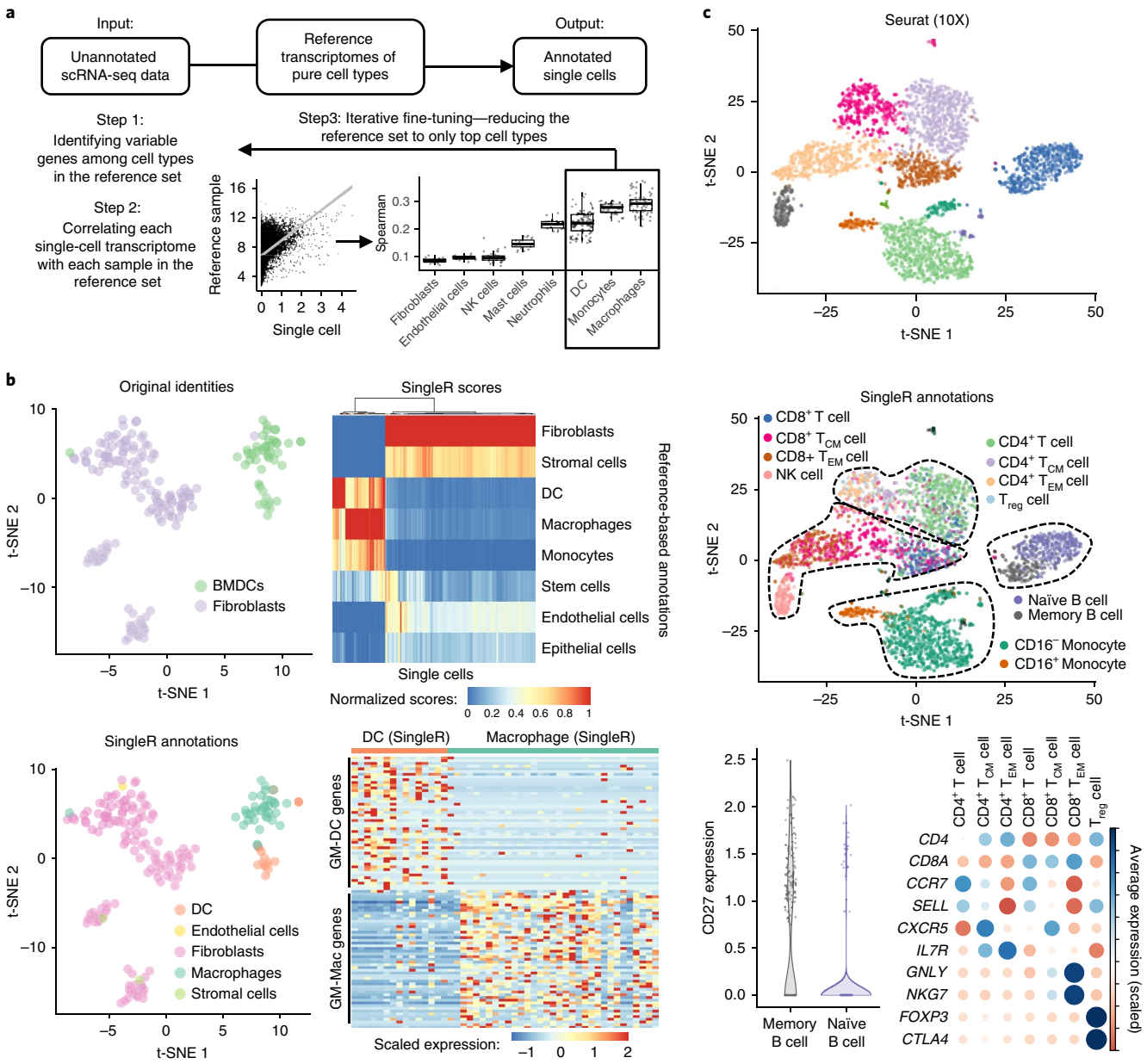

Fig. 1 | Reference-based annotation of scRNA-seq. a, Schematic of SingleR, a protocol for cell type annotation by reference to transcriptomes of pure cell types. **b**, t-SNE plot of a published scRNA-seq data set of fibroblasts and BMDCs[16] (top left). Heatmap of SingleR scores for top correlated cell types (top right). SingleR annotation of cell identity indicated on t-SNE plot (bottom left). Expression in single cells of genes derived by differential expression analysis of published microarray data from bone marrow-derived, GM-CSF-cultured macrophages (GM-Macs) and DCs (GM-DCs)[19] (bottom right). **c**, t-SNE plots of a published scRNA-seq data set of PBMCs[21] analyzed by Seurat without (top) and with (middle) SingleR annotations indicated for individual cells. Cell type markers expressed by SingleR annotation (bottom). DC, dendritic cell; T_{CM}, central memory T cell; T_{EM}, effector memory T cell; T_{reg}, regulatory T cell.

with reference transcriptomic data sets and improved its inferences iteratively (Fig. 1a and Supplementary Note 1). The SingleR pipeline is based on correlating reference bulk transcriptomic data sets of pure cell types with single-cell gene expression. First, we validated SingleR using the Immunological Genome Project (ImmGen) database[12] to annotate mouse scRNA-seq and Encode[13] and Blueprint Epigenomics[14] transcriptomes to annotate human data. In the first step of the SingleR pipeline, a Spearman coefficient was calculated for single-cell gene expression with each of the samples in the reference data set, using only the variable genes in the reference data set. Using only variable genes among the cell types increased the ability to distinguish closely related cell types. This process was performed iteratively using only the top cell types from the previous step and the variable genes among them until only one cell type remained (for details see Methods, Supplementary Note 1, and the SingleR Github repository[15]).

To validate SingleR in mouse, we analyzed a published study of scRNA-seq of mouse bone marrow-derived dendritic cells (BMDC) and fibroblasts[16]. When applied to the ImmGen database[12] as a reference library, SingleR identified 33 of the 48 cells thought in the study to be BMDCs as macrophages (Fig. 1b and Supplementary Note 2). Of note, cells in the study[16] were grown in culture in the presence of the growth factor GM-CSF and then suspended for flow cytometric sorting based on CD11c expression, a protocol[17,18] that has been shown not to result in a pure population of DCs but also results in a proportion of cells bearing macrophage markers[19,20].

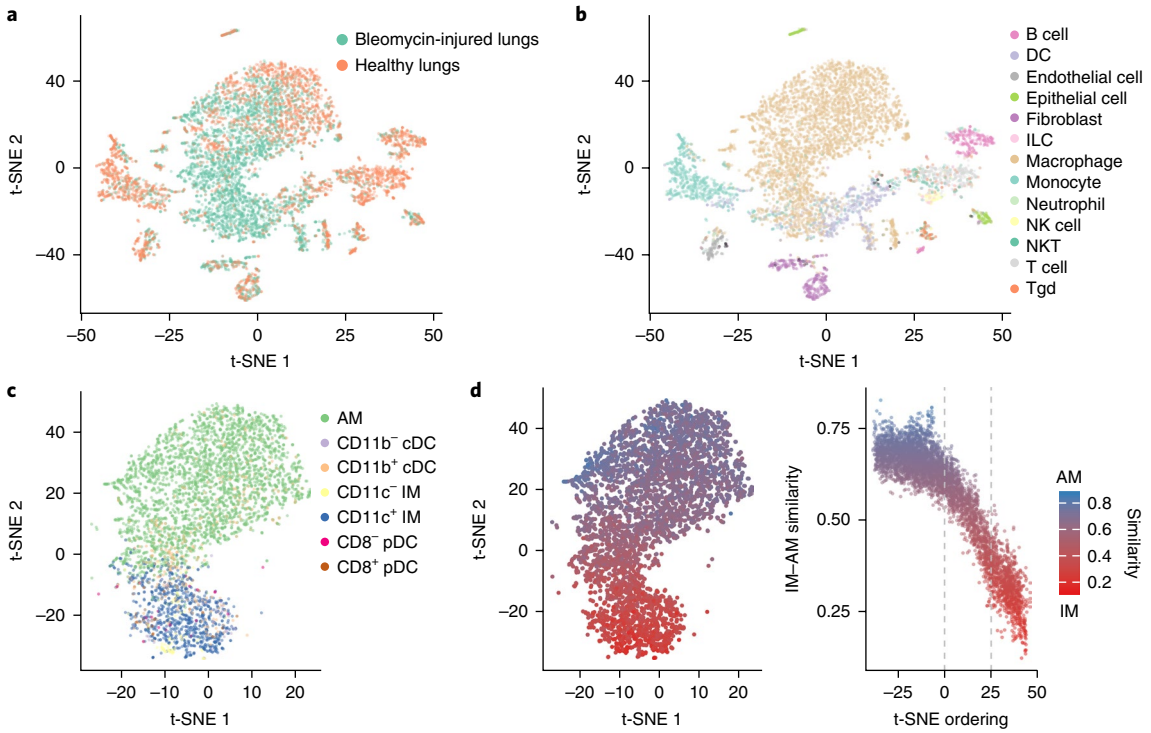

Fig. 2 | Reference data sets reveal heterogeneity of macrophage subtypes in lung fibrosis. a–c, t-SNE plots of single-cell suspensions from mouse whole lung sequenced by Drop-seq color-coded by condition (**a**); annotated by SingleR using the ImmGen database as reference (**b**); and annotated by SingleR with lung-specific myeloid data sets[28,29] (**c**). Data shown pool replicates ($n=3$ biologically independent mice for bleomycin-injured and $n=6$ biologically independent mice for healthy control). **d**, Quantification of similarity in gene expression of individual cells to bulk RNA-seq profiles of alveolar macrophages (AM) and CD11c⁺ interstitial macrophages (IM)[29]. ILC, innate lymphoid cell; NKT, natural killer T cell.

As such, SingleR could make the distinction between macrophages and dendritic cells in an unbiased fashion based on the granularity and breadth of the transcriptomic comparison. In a separate validation, we applied SingleR using the Blueprint and Encode databases as reference comparators for a data set generated by scRNA-seq analysis of human peripheral blood mononuclear cells[21]. In comparison to marker-based and reference-based approaches[21–23], which are limited in their ability to distinguish cell states, SingleR differentiated cellular subtypes such as memory and naïve B and T cells (Fig. 1c and Supplementary Fig. 1). SingleR performed similarly when applied to >50 publicly available scRNA-seq data sets (Supplementary Note 2 and the SingleR web tool[24]). These analyses show that reference-based comparison enhances cell type annotation of scRNA-seq.

SingleR identifies a transitional macrophage in fibrosis. We next applied SingleR to enhance annotation of a lung scRNA-seq data set obtained from whole lungs of adult, male mice (10–12 weeks age) dissociated into single-cell suspension and flow sorted on live cells, using 6 steady-state mice and 3 mice 14 days after a single tracheal instillation of bleomycin, the fibrotic phase of the injury model (Fig. 2a, Supplementary Table 1, and Supplementary Fig. 2). Using the ImmGen database as the reference compendium of data sets for SingleR, we annotated the lung cells and superimposed these cellular identities on the clustering analysis of bleomycin and steady-state cells, pooled together (Fig. 2b and Supplementary Fig. 3). The number of lung macrophages was higher in the bleomycin-treated condition than at steady state (Fig. 2a,b). Because macrophages and dendritic cells have similar transcriptomic profiles[25–27], we repeated SingleR analysis using published RNA-seq data sets of mouse lung macrophages and dendritic cells[28,29] to isolate the macrophages in the pooled bleomycin-injured and steady-state samples (Fig. 2c and Supplementary Fig. 4). SingleR identified two subtypes of macrophages: alveolar and interstitial macrophages (Fig. 2c). The annotation scores for alveolar and interstitial macrophages were strongest at the poles of the macrophage cluster, suggesting a continuum of gene expression between alveolar and interstitial macrophage (Fig. 2d). To quantify this continuum, we performed a similarity analysis[30] by comparison of each cell to mouse RNA-seq data sets of alveolar and interstitial macrophages[29]. This analysis indicated a gradient of cellular identity in the macrophage cluster between alveolar and interstitial macrophages (Fig. 2d).

We next used SingleR annotation scores for each single cell compared with each sample in the reference data set to hierarchically cluster the macrophages. That is, the full range of SingleR correlation scores for each cell with reference to ImmGen was used for unsupervised hierarchical clustering of the macrophages from the lung scRNA-seq data set. This analysis identified three distinct subclusters (C1, C2, and C3), all three detected at varying levels in both steady-state and bleomycin-treated mice (Fig. 3a): cluster C1 was equally present in steady-state lung and bleomycin injury and corresponded to the cells that were annotated as alveolar macrophages by the lung-specific data sets (Fig. 2c); on the other hand, clusters C2 and C3 were enriched in cells isolated from mice with bleomycin-induced fibrosis and expressed *Cx3cr1*, *Ccr2*, *Mafb*, and MHCII genes, consistent with monocytic origin (Fig. 3b and Supplementary Table 2). Notably, 45% of cluster C2 cells (but only 7.3% of cluster C1 cells and 4.7% of cluster C3 cells) expressed at least 33% of cluster C1 genes and at least 33% of cluster C3 genes (Fig. 3c), suggesting that cluster C2 was intermediate in gene expression between C1 and C3. Consistent with this finding, comparison to 2 separate published data sets of lung macrophages[29,31], enabling differential expression analysis between alveolar and interstitial macrophage genes, indicated that C1 had

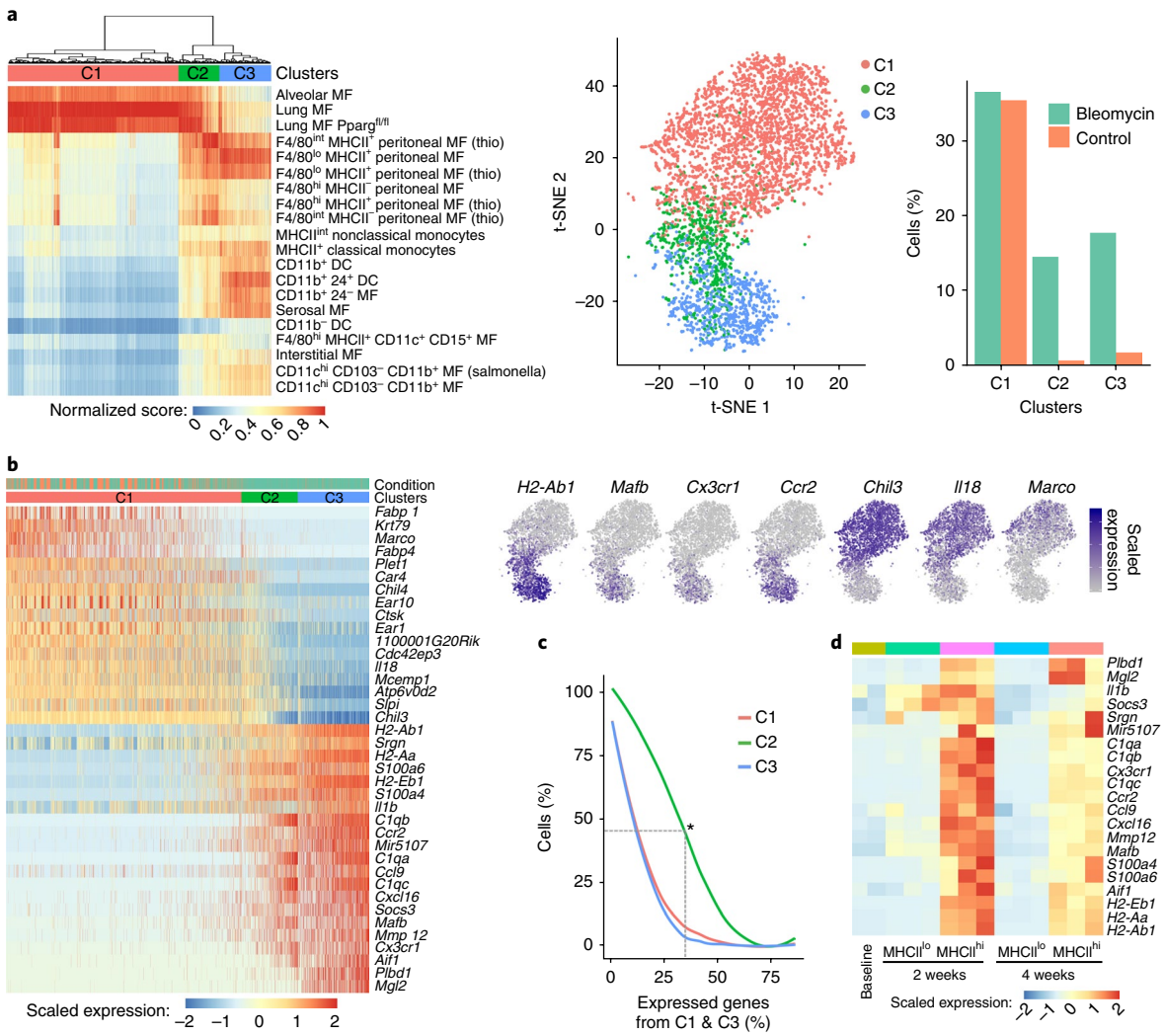

Fig. 3 | Hierarchical clustering by reference annotation reveals a transitional macrophage in fibrosis. a, Heatmap of SingleR annotation scores derived by reference to the ImmGen database with clusters superimposed on the t-SNE plot. MF, macrophages. **b**, Heatmap of genes differentially expressed between C1 and C3 with examples superimposed on the t-SNE plot. Cells are ordered by the number of non-zero C3 genes. **c**, Percentage of cells in each cluster that express genes in **b** (differentially expressed between C1 and C3). Asterisk indicates that 45% of C2 cells express at least 33% of C1 genes and at least 33% of C3 genes. **d**, Heatmap of the expression of C3 genes in bulk RNA-seq of alveolar macrophages (SiglecF$^+$CD11c$^+$) sorted on MHCII^{lo} at baseline and both MHCII^{lo} and MHCII^{hi} at two time points after bleomycin lung injury ($n = 2$ biologically independent mice for baseline, $n = 3$ biologically independent mice at 2 weeks, and $n = 3$ biologically independent mice at 4 weeks).

higher expression of alveolar macrophage genes than cluster C2, which in turn had higher expression of alveolar macrophage genes than cluster C3 (Supplementary Fig. 5).

To determine the directionality of the transitional state of C2 cells across time in the model, we performed bulk RNA-seq of SiglecF$^+$CD11c$^+$MHCII^{lo} and SiglecF$^+$CD11c$^+$MHCII^{hi} cells, paralleling clusters C1 and C2, respectively. These cells were sorted from lungs of bleomycin-injured, 10–12-week-old, wild-type male mice. SiglecF$^+$CD11c$^+$MHCII^{hi} macrophages had higher expression of genes identified by scRNA-seq in cells from clusters C2 and C3 compared with SiglecF$^+$CD11c$^+$MHCII^{lo} cells at 2 weeks after injury (Fig. 3d and Supplementary Fig. 6a). However, the expression of these genes in SiglecF$^+$CD11c$^+$MHCII^{hi} cells at 4 weeks after injury was reduced to levels more similar to those detected in SiglecF$^+$CD11c$^+$ alveolar macrophages at steady state (Fig. 3d). These data suggest that between the 2- and 4-week time points, SiglecF$^+$CD11c$^+$MHCII^{hi} cells, which correspond to cluster C2 identified by SingleR, transition toward a steady-state alveolar macrophage profile (corresponding to cluster C1).

Cluster C2 and C3 markers are found in alternate lung fibrosis models. To test the relevance of the identified gene expression pattern to other models of fibrosis, we used a mouse model of lung fibrosis induced by epithelial telomere dysfunction. In this model, *SPC-CreERT2* mice are crossed with a floxed *Trf1* allele for tamoxifen-inducible deletion of Trf1, a shelterin complex protein that protects telomeres. Trf1 deletion after tamoxifen induction leads to age-dependent telomere shortening and is associated with cellular senescence and progressive fibrosis through the lifespan[32]. Whole-lung lysates from *SPC-CreERT2/Trf1*$^{fl/fl}$ mice harvested at 3 and 9 months after initiation of tamoxifen treatment and *Trf1*$^{fl/fl}$ control mice harvested at 9 months after tamoxifen were submitted for gene expression profiling by microarray. We detected a progressive increase in the expression of genes characteristic of clusters C2 and C3 in the lung lysates from tamoxifen-treated *SPC-CreERT2/Trf1*$^{fl/fl}$ mice compared with control, with peak expression at 9 months (Fig. 4a), coincident with peak fibrosis in this model. Although these data represent whole-lung rather than isolated macrophages, together with the known increase in macrophages during the fibrotic period

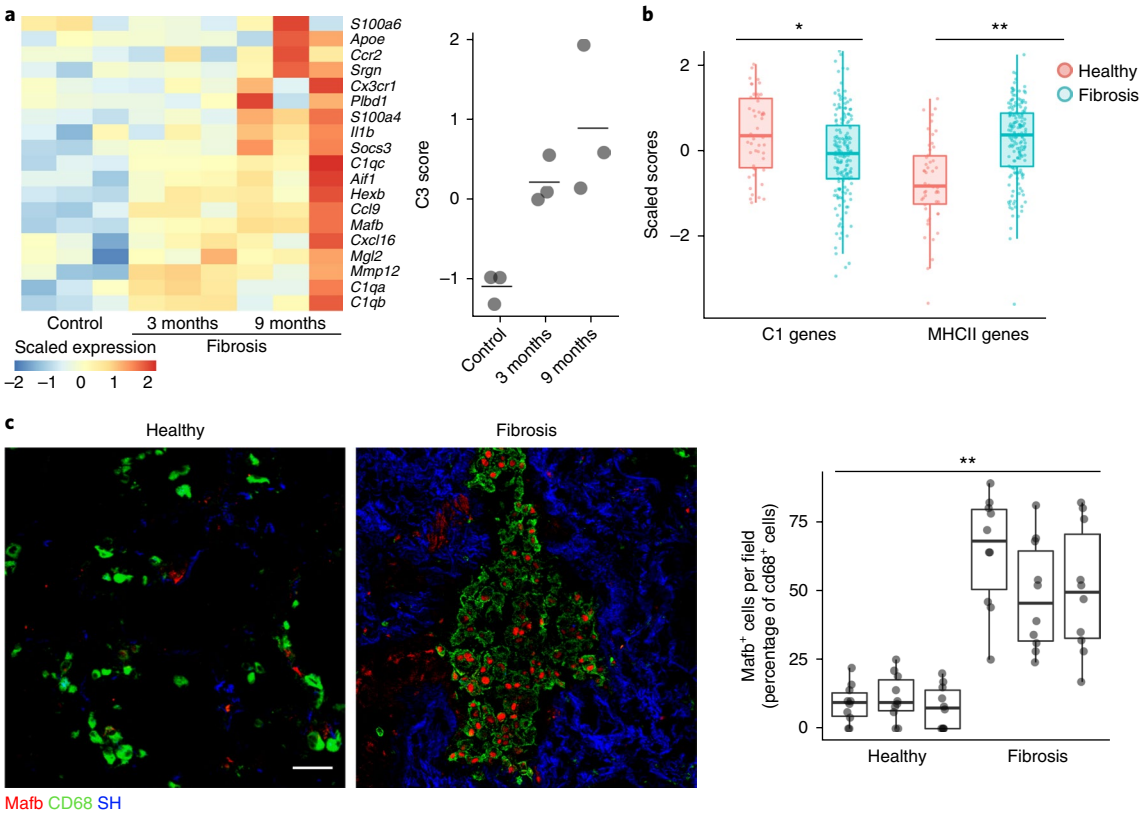

Fig. 4 | Markers of macrophages identified by scRNA-seq are found in multiple fibrosis models. a, Heatmap of C3 genes from bleomycin-induced fibrosis measured by microarray in an epithelial telomere dysfunction model of progressive lung fibrosis ($n=3$ biologically independent mice in each condition, center value is mean). **b**, Gene set enrichment analysis scores of human orthologs of cluster C1 genes and of MHCII genes in bulk RNA-seq samples from patient lung biopsy specimens[34]. Points indicate individual microarray samples from $n=167$ and $n=50$ patients and controls, respectively. **c**, Immunofluorescence of human fibrotic and healthy control lung (representative images are shown; quantitation is for $n=3$ independent fibrotic patient lung and $n=3$ independent healthy lung samples, 10 images per sample). Scale bar, 50 μm. SH, second harmonic imaging of collagen. Box plot center line is median, box limits are upper and lower quartiles, and whiskers denote the largest and smallest values no more than 1.5 times the interquartile range from the limits. Wilcoxon test two-tailed P values are presented. *$P<0.01$; **$P<0.0001$.

in this model[32], the results are consistent with a common genetic signature and a generalizable pattern of macrophage response during fibrogenesis.

We then asked whether the gene expression profiles identified in mouse were relevant to human lung fibrosis. To address this question, we performed single-sample gene set enrichment analysis[33] to quantify the expression of human orthologs of genes of interest in a published microarray data set of 167 lung samples from patients with lung fibrosis and 50 healthy controls[34]. This analysis showed a significant decrease in the expression of genes associated with alveolar macrophages (cluster C1) and a significant increase in the expression of MHC II genes (cluster C2 and C3) in the human fibrosis samples compared with the healthy controls (Fig. 4b). Moreover, immunofluorescence of lung tissue indicated the expression of MAFB, a marker detected specifically in clusters C2 and C3, in CD68$^+$ macrophages from patients with idiopathic lung fibrosis compared with healthy controls (Fig. 4c). Taken together, these studies support the relevance of the C2 and C3 markers identified in the bleomycin model to multiple contexts of fibrosis, mouse and human.

Cluster C2 macrophages localize to the fibrotic niche. Next, we tried to identify the anatomic niche of macrophage clusters C1, C2, and C3. Because *Cx3cr1* was expressed in clusters C2 and C3 in the bleomycin-treated mice, we used mice expressing the tamoxifen-inducible allele *Cx3cr1-CreERT2*[35], which were crossed with *Rosa26-loxp-STOP-loxp-TdTomato* reporter mice to mark *Cx3cr1*-expressing cells with the fluorophore TdTomato. Tamoxifen was given every other day starting 1 day before bleomycin treatment until lung analysis on days 7 or 14, while control mice received the same tamoxifen schedule through day 14 without bleomycin to determine the baseline levels of TdTomato$^+$ cells. At baseline, TdTomato$^+$ cells represented 6% of the live lung cells by flow cytometry, probably representing the interstitial macrophages present in steady-state lung (Fig. 5a and Supplementary Fig. 6b). In bleomycin-treated mice we observed an increase in the percentage of TdTomato$^+$ cells among total lung live cells by flow cytometry at both day 7 (23%) and day 14 (38%) compared with baseline (Fig. 5a). Of note, the expression of the alveolar macrophage marker SiglecF on TdTomato$^+$ cells increased (as assessed by mean fluorescence intensity) across the time series (Fig. 5a), indicating the acquisition of an alveolar macrophage marker by monocyte-derived macrophages across time.

We next performed lung immunofluorescence analysis of *Cx3cr1-CreERT2/Rosa26-loxp-STOP-loxp-TdTomato* mice at day 14 after bleomycin injury to determine the spatial organization of the macrophages in the fibrotic lung. TdTomato$^+$SiglecF$^+$ macrophages were detected in direct contact with clusters of fibroblasts as defined by areas delimited by multiple adjacent Pdgfrb$^+$ cells (Fig. 5b). On the other hand, TdTomato$^-$SiglecF$^+$ alveolar macrophages (cluster C1) were detectable throughout the lung except in regions where Pdgfrb$^+$ fibroblasts accumulated at sites of injury (Fig. 5b). The fact

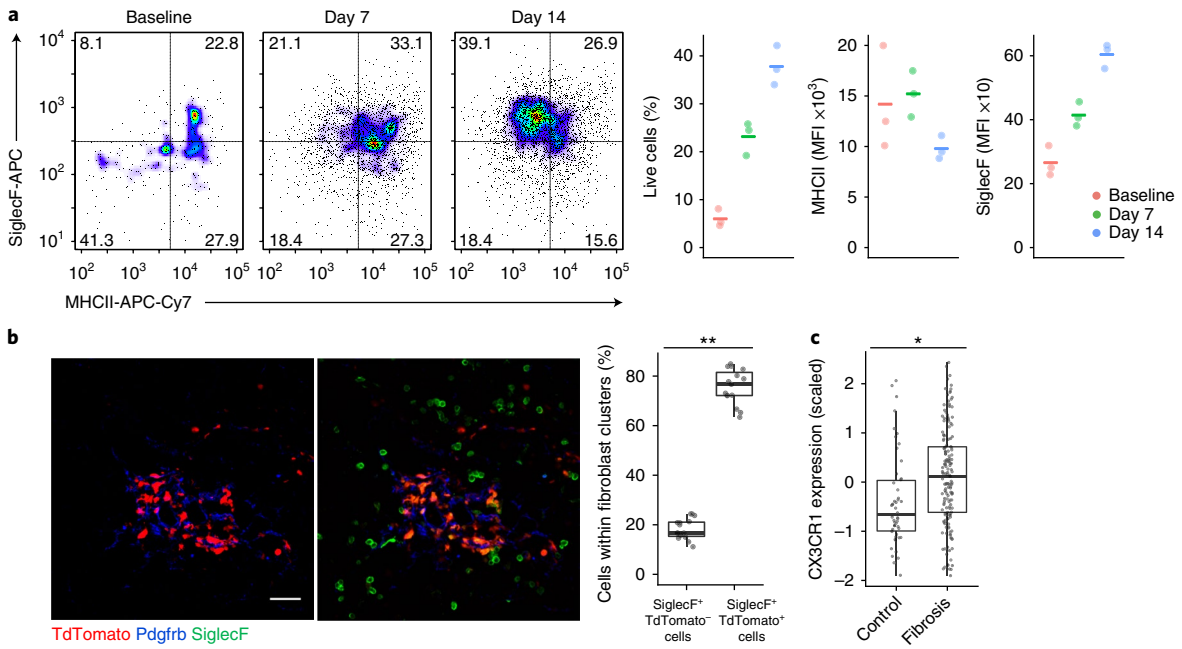

Fig. 5 | Transitional macrophages expressing CX3CR1, MHCII, and SiglecF localize to sites of fibroblast accumulation after lung injury. a, Flow cytometry of TdTomato⁺ lung cells in *Cx3cr1*-CreERT2/Rosa26-loxp-STOP-loxp-TdTomato mice with tamoxifen administration before and after injury followed by flow cytometry, with representative pseudocolor plots and values plotted for TdTomato⁺ cells at three different time points ($n = 3$ biologically independent mice per group; mean is marked). **b**, Lung immunofluorescence at 14 d after injury in *Cx3cr1*-CreERT2/Rosa26-loxp-STOP-loxp-TdTomato mice treated as in **a** (a representative field of view with and without siglecF immunofluorescence is shown; quantitation is for $n = 3$ biologically independent mice, 5 images per mouse). Scale bar, 50 μm. **c**, Expression of *CX3CR1* in bulk RNA-seq samples from patient lung biopsy specimens[34]. Box plot center lines are median, box limits are upper and lower quartiles, and whiskers denote the largest and smallest values no more than 1.5 times the interquartile range from the limits. Wilcoxon test two-sided *P* values are presented. **P* <0.001; ***P* <0.0001.

that TdTomato⁺ cells within the scar were also SiglecF⁺ suggests that the macrophages localizing to fibrotic zones were in a transitional state representative of cluster C2. Human *CX3CR1* mRNA was also more highly expressed in lung microarray samples from 167 patients with lung fibrosis than in samples from 50 healthy controls (Fig. 5c)[34], underscoring the relevance of the C2 cluster for human disease. Taken together, these results show that macrophage cluster C2 localizes to the fibrotic niche.

CX3CR1-expressing macrophages drive lung fibrosis. Because of the proximity of transitional macrophages to fibroblasts at fibrotic sites, we investigated whether the CX3CR1⁺ macrophages provided trophic or mitogenic support to fibroblasts. Macrophage-derived Pdgf has been shown to support fibroblast growth in co-culture[36]. In the lung scRNA-seq data set, *Pdgfa* was uniquely expressed in the fibrosis-associated clusters C2 and C3 (Fig. 6a). Furthermore, *Pdgfa* mRNA was more highly expressed by bulk RNA-seq in SiglecF⁺CD11c⁺MHCII^{hi} (cluster C2) macrophages than SiglecF⁺CD11c⁺MHCII^{lo} macrophages (cluster C1) at 2 weeks (Fig. 6a). Lung immunofluorescence analysis of *Cx3cr1-CreERT2/Rosa26-loxp-STOP-loxp-TdTomato* mice at day 14 after bleomycin injury indicated that the homodimeric form of Pdgfa, Pdgf-aa, was highly and uniquely expressed in TdTomato⁺ cells adjacent to fibroblasts that were positive for expression of Pdgfra, the receptor for Pdgf-aa (Fig. 6b).

To test whether cluster C2 cells supported fibroblast migration or proliferation through production of Pdgf-aa, we performed in vitro assays with 3T3 fibroblasts, an embryonic mouse fibroblast cell line. Conditioned media from SiglecF⁺CD11c⁺MHCII^{hi} macrophages (corresponding to cluster C2) sorted from wild-type mice 14 days after bleomycin injury and cultured for 2 days enhanced gap closure by 3T3 fibroblasts compared with conditioned media from SiglecF⁺CD11c⁺MHCII^{lo} macrophages (corresponding to cluster C1) isolated from the same mice (Fig. 6c). Antibody blockade of Pdgf-aa inhibited the enhanced gap closure seen in SiglecF⁺CD11c⁺MHCII^{hi} compared with SiglecF⁺CD11c⁺MHCII^{lo} conditioned media (Fig. 6c), suggesting that the fibroblast response was dependent on Pdgf-aa secreted by SiglecF⁺CD11c⁺MHC^{hi} macrophages. In a second assay, when TdTomato⁺ cells sorted 14 days after bleomycin injury from tamoxifen-induced *Cx3cr1-CreERT2/Rosa26-loxp-STOP-loxp-TdTomato* mice were co-cultured with 3T3 fibroblasts, Pdgf-aa blocking antibody decreased the proliferation of fibroblasts as measured by 5-ethynyl-2′-deoxyuridine (EdU) uptake (Fig. 6d).

Analysis of the lung scRNA-seq data set including cells from both bleomycin-treated and steady-state lung pooled together and annotated by SingleR showed three clusters of fibroblasts, cluster A, B, and C (Fig. 6e). Pdgfra was not detected in the lung scRNA-seq data set, which may reflect the limitation of the depth of sequencing with the single-cell method; however, expansion of Pdgfra⁺ fibroblasts after bleomycin treatment has been reported in a study analyzing the fibroblast compartment in this model of fibrosis[37]. Next, we used gene set analysis[38] to quantify cell cycle genes in the lungs of bleomycin-treated mice compared with steady state 14 days after bleomycin treatment. Cluster A showed the highest expression of cell cycle genes and was predominantly composed of cells from bleomycin-treated mice (Fig. 6e).

To investigate the role of CX3CR1⁺ cells in the proliferation of fibroblasts in vivo, *Cx3cr1-CreERT2/Rosa26-loxp-STOP-loxp-Diphteria Toxin A* mice were treated with tamoxifen between day 8 and day 21 after bleomycin injury to induce diphtheria toxin A-dependent deletion of CX3CR1⁺ cells during the fibrotic phase of bleomycin lung injury. At 21 days after bleomycin injury, bleomycin-injured, tamoxifen-induced *Cx3cr1-CreERT2/Rosa26-*

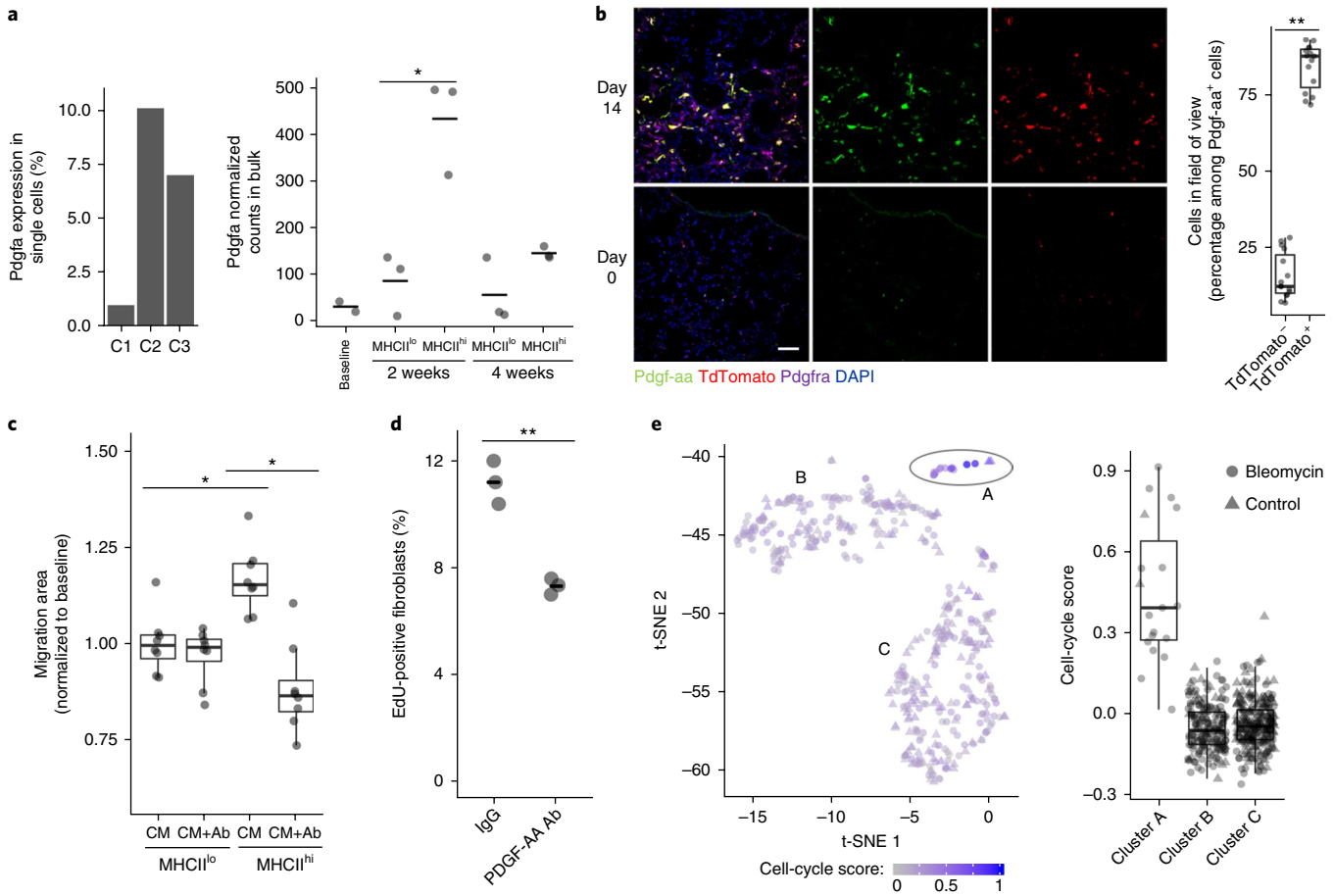

Fig. 6 | Pdgf-aa is a macrophage-derived trophic factor for fibroblasts. a, *Pdgfa* expression in lung scRNA-seq macrophage clusters and by bulk RNA-seq of SiglecF$^+$CD11c$^+$ lung macrophages after bleomycin injury. Mean is marked. Wald test 2-sided *P* value is presented (for the lung scRNA-seq data, $n=3$ biologically independent mice for bleomycin-injured and $n=6$ biologically independent mice for healthy control; for the bulk RNA-seq data, $n=2$ biologically independent mice for baseline, $n=3$ biologically independent mice for all other conditions). **b**, Lung immunofluorescence at baseline (bottom) and 14 d after injury (top) in tamoxifen-induced *Cx3cr1*-CreERT2/Rosa26$_{loxp}$STOP$_{loxp}$-TdTomato mice (each panel of three micrographs shows a representative field of view, the leftmost image a merge, the center Pdgf-aa alone, and the rightmost TdTomato alone; quantitation is for the 14-d time point in $n=3$ biologically independent mice, 5 images per mouse). Scale bar, 50 µm. **c**, 3T3 fibroblast gap closure in response to conditioned media (CM) from lung macrophages sorted by MHCII expression, with and without Pdgf-aa blocking antibody (Ab; $n=8$ biologically independent mice total, each point a separate assay). **d**, EdU quantitation in 3T3 fibroblasts co-cultured with TdTomato$^+$ cells sorted from tamoxifen-induced *Cx3cr1*-CreERT2/Rosa26-loxp-STOP-loxp-TdTomato mice 14 d after injury. Mean is marked ($n=3$ biologically independent co-cultures). **e**, t-SNE of SingleR-annotated fibroblasts from the lung scRNA-seq data set showing intensity of cell cycle signature ($n=3$ biologically independent mice for bleomycin, $n=6$ biologically independent mice for control). Box plot center lines are median, box limits are upper and lower quartiles, and whiskers denote the largest and smallest values no more than 1.5 times the interquartile range from the limits. Wilcoxon test two-sided *P* values are presented. *$P<0.01$; **$P<0.0001$.

loxp-STOP-loxp-Diphteria Toxin A mice had significantly fewer macrophages quantified by immunofluorescent labeling of either Mertk$^+$ or SiglecF$^+$ cells in the lung (Supplementary Fig. 7), significantly decreased numbers of cells expressing the fibroblast markers Pdgfra$^+$ and Pdgfrb$^+$ (Fig. 7a), and significantly decreased lung collagen as measured by hydroxyproline assay compared with bleomycin-injured wild-type mice (Fig. 7b). This result indicates that CX3CR1$^+$ macrophages were required for the expansion of the fibroblast compartment and the deposition of fibrotic scar.

In human patients, one pathologic feature of idiopathic pulmonary fibrosis is the presence of normal tissue adjacent to regions of fibrotic foci, which are commonly found in the subpleural region of the lung[39]. In three lung explants displaying such regional heterogeneity, we found that PDGF-AA$^+$ macrophages were associated with the fibrotic rather than non-fibrotic regions (Fig. 7c). Together, these results suggest the dependence of fibroblast proliferation, which is a key step in fibrogenesis[40], on trophism exerted by PDGF-AA$^+$ macrophages recruited to sites of injury.

Discussion

We show here that during fibrogenesis CX3CR1$^+$SiglecF$^+$ macrophages localize to sites of Pdgfra$^+$ and Pdgfrb$^+$ fibroblast accumulation after bleomycin-induced injury in the lung. Deletion of these macrophages during the fibrotic period (days 8–21) eliminated Pdgfra$^+$ or Pdgfrb$^+$ fibroblasts and decreased total lung collagen. Mechanistic studies indicated that fibroblast migration and proliferation were dependent on secretion of Pdgf-aa by SiglecF$^+$CD11c$^+$MHCII^{hi} and CX3CR1$^+$ cells, respectively, suggesting that paracrine interactions between these macrophages and fibroblasts sustain the fibroblast compartment and tissue fibrosis.

Hierarchical clustering of correlation scores for macrophages with bulk data sets from ImmGen enabled the deconvolution of macrophage heterogeneity and identified three clusters of lung macrophages: alveolar macrophages (cluster C1), interstitial macrophages (cluster C3), and an intermediate cluster C2 bearing likeness to both alveolar and interstitial macrophages by gene expression. This analysis motivated the hypothesis that C2 is a transitional state

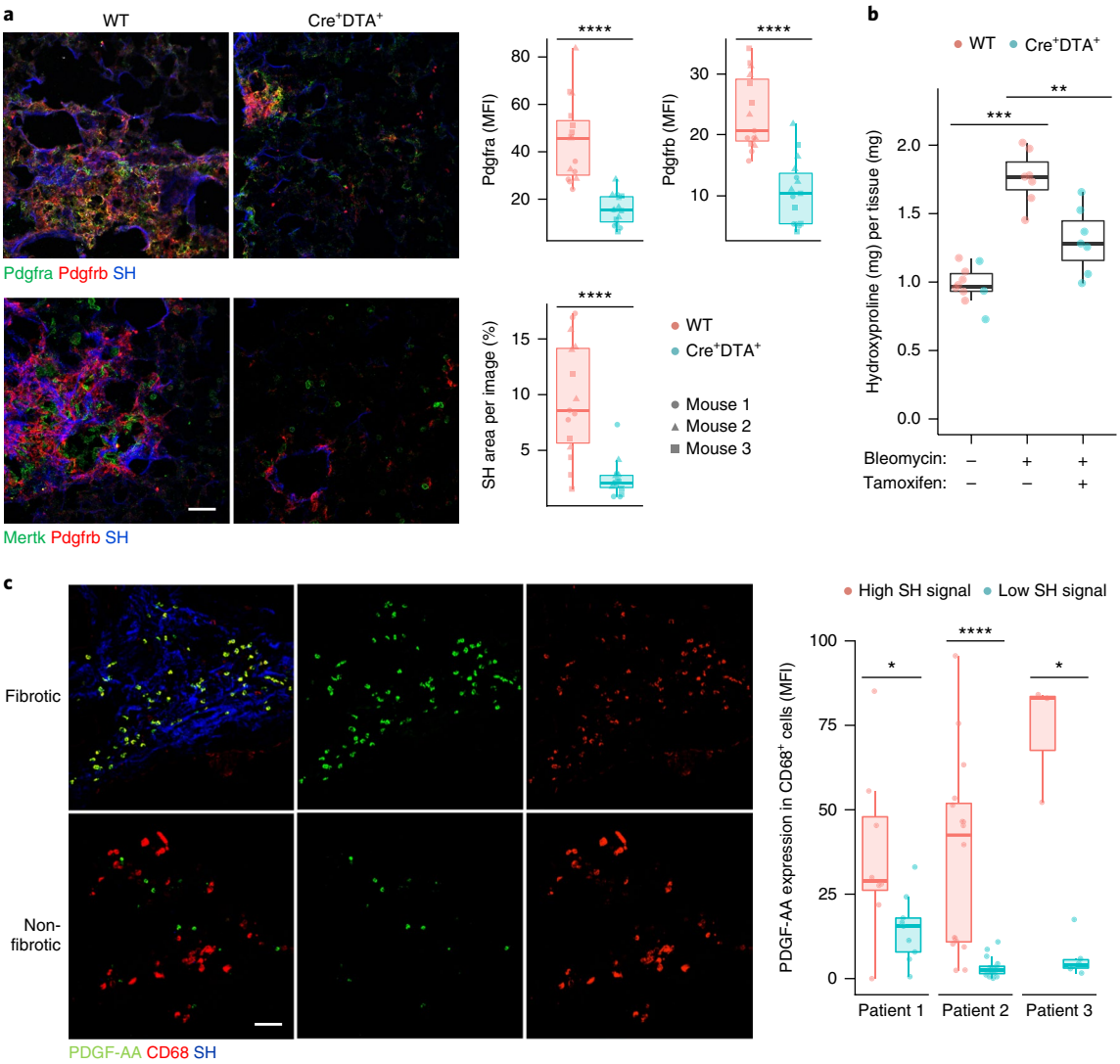

Fig. 7 | *Cx3cr1*-derived macrophages drive fibroblast accumulation and fibrosis after lung injury. a, Lung immunofluorescence of fibroblast and macrophage markers 21 d after bleomycin injury in *Cx3cr1-CreERT2/Rosa26-loxp-STOP-loxp-Diphtheria Toxin A* mice treated with tamoxifen between day 8 and day 21 and in wild-type mice (representative images are shown; quantitation is for $n=3$ biologically independent mice per group, 5 images per mouse). WT, wild type; MFI, mean fluorescence intensity. SH, second harmonic imaging of collagen. **b**, Fibrosis measured by hydroxyproline assay for lung collagen content in mice treated as in **a** (blue points are *Cx3cr1-CreERT2/Rosa26*$_{loxp}$*STOP*$_{loxp}$*-Diptheria Toxin A* mice and red points are wild type; $n=10$, $n=7$, and $n=7$ biologically independent mice per group from left to right). Scale bar, 50 μm. **c**, Immunofluorescence of fibrotic (high SH) and non-fibrotic (low SH) regions from explanted human fibrotic lung (each panel of three micrographs shows a representative field of view, the leftmost image a merge, the center PDGF-AA alone, and the rightmost CD68 alone), with quantification of individual images from 3 separate patients (8 high and 9 low regions in patient 1, 16 high and 16 low in patient 2, and 3 high and 6 low in patient 3). Scale bar, 50 μm. Box plot center lines are median, box limits are upper and lower quartiles, and whiskers denote the largest and smallest values no more than 1.5 times the interquartile range from the limits. Wilcoxon test two-sided *P* values are presented. *$P<0.05$; **$P<0.01$; ***$P<0.001$; ****$P<0.0001$.

arising after injury, which was supported by labeling and expression studies revealing kinetic transition toward increasing expression of the alveolar macrophage (C1) marker SiglecF by cells expressing the C2 and C3 marker *Cx3cr1*.

In cell culture systems, we found that antibody blockade of Pdgf-aa decreased the enhanced 3T3 fibroblast gap closure induced by conditioned media from MHCII^{hi} alveolar macrophages (corresponding to cluster C2) as well as the proliferation of 3T3 fibroblasts in co-culture with CX3CR1^{+} macrophages from bleomycin-injured mice. Of note, one of the few Food and Drug Administration-approved therapies for lung fibrosis, the small molecule receptor tyrosine kinase inhibitor nintedanib, blocks PDGF receptor signaling[41], and Pdgfra expressed by fibroblasts drives fibrosis in multiple organs in mice[42–47], including lung fibrosis induced by bleomycin treatment[48]. These observations indicate that the Pdgf pathway is crucial to disease pathogenesis. Our results implicate the CX3CR1^{+}SiglecF^{+} macrophage as the source of Pdgf-aa in the fibrotic niche.

A recent study reported an increase in the number of Ceacam1^{+}Msr1^{+}Ly6C^{-}F4/80^{-}Mac1^{+} monocytes, named segregated-nucleus-containing atypical monocytes or SatM, in the bronchoalveolar lavage at day 13 after bleomycin treatment in mice[11]. SatM cells were found to be pro-fibrotic when transferred to the lung intravenously. However, macrophages were not tested, whereas our study characterizes these macrophage populations specifically. Furthermore, cells in that study were isolated by bronchoalveolar lavage, which does not efficiently recover cells from fibrotic regions. In the current study, we used dissociated whole lung for single-cell

analysis, which enabled an unbiased view of all cellular subtypes in the lung.

Our observation that ablation of macrophages expressing *Cx3cr1* decreased fibroblasts and lung collagen content after bleomycin treatment has relevance for human disease. A similar macrophage gene expression profile was identified in an alternative mouse model of lung fibrosis mediated by telomere dysfunction, and analysis of human lung samples indicated increased expression of *CX3CR1* and MHCII genes in patients with fibrosis as well as the expression of the C2 and C3 markers MAFB and PDGF-AA in regions of active fibrosis. These findings underscore the potential therapeutic significance of our functional studies in mice.

A limitation of using *Cx3cr1-CreERT2* is that *Cx3cr1* is expressed in both clusters C2 and C3, limiting the specificity of our studies. Our analysis suggests an evolution of cluster C3 towards cluster C1, with C2 representing a transitional state. Future studies should characterize the kinetics of this transition in greater depth and determine whether specific ablation of either cluster across time may be possible. Second, our annotation of cell states using SingleR relies on the particularities of the reference data sets. The ImmGen database is a large compendium of cell types from multiple organs, which greatly facilitated our clustering analysis by differential annotation. Nonetheless, we anticipate that the power of SingleR to identify new cell types or cell states will continue to increase as single-cell and bulk RNA-seq data sets are amassed from a range of contexts.

In summary, this study identified a profibrotic macrophage subpopulation that localizes to sites of fibrotic scar with activating effects on the mesenchyme. Furthermore, the identification of markers specific for the profibrotic macrophages, including CX3CR1 and PDGF-AA, could be useful for therapeutic targeting in clinical fibrosis. Our results also demonstrate the value of using reference-based annotation of single-cell data sets to map the heterogeneity of canonical cell types, such as macrophages, and to enhance the differentiation of transcriptomically unique subclusters within the data. Coupled with lineage tracing and functional studies, this approach should serve as a broadly applicable platform for study of specific cellular compartments in health and disease.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, statements of data availability and associated accession codes are available at https://doi.org/10.1038/s41590-018-0276-y

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Author contributions

D.A. developed SingleR and performed computational analysis of single-cell data under the guidance of A.J.B. A.P.L. performed in vivo and in vitro experiments under the guidance of M.B. and with the assistance of E.W. and S.C. L.L. performed microfluidic capture of single-cell transcriptomes, library preparation, and sequencing under the guidance of A.R.A. V.F., A.H., and E.W. prepared breeding and experimental stocks of genetically modified mice and performed lung injury models under the guidance of M.B. P.J.W. contributed acquisition, storage, and processing of human samples and, with R.P.N., acquired lung microarray data from mice with telomere dysfunction. D.A. prepared the figures. M.B. conceived of the work, supervised experimental planning and execution, and wrote the manuscript with input from D.A., A.P.L., and L.L.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to M.B.

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Methods

Mice. Ai14 (*Rosa26-LSL-tdTomato*), DTA (*Gt(ROSA)26Sor*$^{tm1(DTA)Lky}$), *CX3CR1-CreERT2* (*Cx3cr1*$^{tm2.1(cre/ERT2)Jung}$), and *TRF1*$^{fl/fl}$, all on C57BL/6 background, wild-type C57BL/6, and wild-type 129S1 mice were obtained from the Jackson Laboratory. SPC-CreERT2rtTA mice, on C57BL/6 background, were obtained from Hal Chapman. Strain-appropriate controls were used in all experiments, and genotyping of the mice was performed by PCR. *Cx3cr1-CreERT2* mice were administered 2 mg tamoxifen (Sigma) via intraperitoneal injection every other day for Cre induction. Mice were maintained in specific-pathogen-free conditions in the Animal Barrier Facility of the University of California, San Francisco. All animal experiments were in accordance with protocols approved by the University of California, San Francisco Institutional Animal Care and Use Committee.

Microfabrication of devices. The poly(dimethylsiloxane) (PDMS) Drop-seq droplet microfluidic device is fabricated with standard soft lithography techniques. Photoresist masters are created by spinning a layer of photoresist SU-8 (Microchem) onto a 7.5-cm silicon wafer (University Wafer), then baking at 95 °C for 20 min. Then, the photoresist is subjected to a 3-min ultraviolet light exposure over Drop-seq[38] photolithography masks (CAD/Art Services). After ultraviolet exposure, the wafers are baked at 95 °C for 10 min, developed in fresh propylene glycol monomethyl ether acetate (Sigma Aldrich), rinsed with fresh propylene glycol monomethyl ether acetate, and baked at 95 °C for 1 min to remove solvent. The microfluidic devices are fabricated by curing PDMS (10:1 polymer-to-crosslinker ratio) over the photoresist master[49]. The devices are cured in a 65-°C oven for 2 h and extracted with a scalpel, and inlet ports are added using a 0.75-mm biopsy core (World Precision Instruments). The device is bonded to a glass slide using O_2 plasma treatment, and channels are treated with Aquapel (PPG Industries) to render them hydrophobic. Finally, the devices are baked at 65 °C for 20 min to dry the Aquapel before they are ready for use.

Drop-seq and data analysis. For scRNA-seq analysis of lung fibrosis, bleomycin (3 U kg^{-1}; Hospira) or water was instilled intratracheally to anesthetized male 129S1 mice, age 10–12 weeks. After 14 d, mice were killed, and lungs were perfused with PBS and dissociated to a single-cell suspension in Roswell Park Memorial Institute RPMI medium containing 0.13 U Liberase TM (Roche) and dissociated using gentleMACS (Miltenyi Biotec). Cells were passed through 70-µm and 40-µm strainers followed by FACS sorting of 4,6-diamidino-2-phenylindole (live) cells. The single-cell RNA-seq experiment is performed based on the Drop-seq protocol[38]. Briefly, the barcoded Drop-seq beads (MACOSKO-2011-10, ChemGenes) and single-cell suspensions from dissociated mouse lung are re-suspended to 100 beads µl^{-1} in PBS-BSA buffer and 120 cells µl^{-1} in Drop-seq lysis buffer (with additional 1 M NaCl added), respectively. Monodisperse droplets ~1 nl in size were generated using the fabricated Drop-seq device. We used HFE7500 with 2% w/v ionic krytox as oil phase. The collected droplets were broken with perfluorooctanol (Sigma) in 30 ml 6× SSC buffer. The beads were then washed and re-suspended in a reverse transcriptase mix for reverse transcription and the template switch reaction. Exonuclease I is used to remove unextended primers following the reverse transcription reaction. The beads were then washed, counted, aliquoted into PCR tubes at 6,000 beads per PCR reaction, and PCR amplified for 17 cycles: 95 °C for 3 min; then 4 cycles of: 98 °C for 20 sec, 65 °C for 45 sec, 72 °C for 3 min; 13 cycles of: 98 °C for 20 sec, 67 °C for 20 sec, 72 °C for 3 min; and, finally, 72 °C for 5 min. The PCR reactions from one Drop-seq run were pooled and purified with AMPure XP beads; the amplified complementary DNA was quantified with Qubit dsDNA High-Sensitivity Assay and checked on a BioAnalyzer High Sensitivity Chip (Agilent). The cDNA was fragmented and amplified for sequencing with the Nextera XT DNA Sample Prep Kit (Illumina) using a primer that enabled the specific amplification of only the 3′ ends and Illumina index primers N70X. The libraries were purified with Ampure beads, quantified with Quabit dsDNA High-Sensitivity Assay, checked on a BioAnalyzer High Sensitivity Chip, and then sequenced on Illumina Miseq or Hiseq2500. Paired-end sequence reads were processed mostly as previously described[38]. Briefly, cell barcode and unique molecular identifier are extracted from Read 1. Read 2 was aligned to the mouse mm10 genome (UCSC) using Bowtie[50]. Reads mapping to exonic regions of genes of mouse mm10 genome were recorded. The number of transcripts for a given gene within a cell barcode was determined by counting unique molecular identifiers and they were compiled into a digital gene expression matrix.

For initial clustering, we used Seurat v2.2[38] to cluster pooled data counts from 9 batches (42,000 cells, estimated based on the number of Drop-seq beads recovered, the concentration of cells, and droplet size). Cells with less than 500 non-zero genes were filtered out; genes expressed in only 1 cell were omitted. Altogether, the filtered data contained 8,366 cells and 13,861 genes (Supplementary Table 1). Expression data were log-normalized, and data were scaled to regress out differences in number of detected molecules. Variable genes were identified using the FindVariableGenes function with x.low.cutoff = 0.0125, x.high.cutoff = 3, and y.cutoff = 0.5. The t-distributed stochastic neighbor embedding (t-SNE) plot was computed using the top 10 principal components.

Cell type annotation of single cells (SingleR). The SingleR pipeline is based on correlating gene expression of pure cell types with single-cell gene expression. SingleR can be used with any reference data set. For mouse samples, we used the ImmGen database, a collection of 830 microarray samples, which we classified to 20 main cell types and further annotated to 253 subtypes[12]; a data set of 358 mouse RNA-seq samples annotated to 28 cell types collected, processed, and shared, courtesy of Bérénice Benayoun, can also be used. For human samples we used the data sets from Blueprint Epigenomics, 144 RNA-seq pure immune samples annotated to 28 cell types[14], and Encode, 115 RNA-seq pure stroma and immune samples annotated to 17 cell types[13]. More reference data sets for mouse and human are available with the SingleR R package. For specific applications, smaller data sets can be applicable (as demonstrated in Fig. 2c).

The annotation in *SingleR* is performed for each single cell independently. First, a Spearman coefficient is calculated for single-cell expression with each of the samples in the reference data set. The correlation analysis is performed only on variable genes in the reference data set. Next, multiple correlation coefficients per cell type according to the named annotations of the reference data set are aggregated to provide a single value per cell type per single cell. SingleR uses the 80th percentile of correlation values, to prevent misclassification due to heterogeneity in the reference samples. Finally, in the fine-tuning step, SingleR reruns the correlation analysis, but only for the top cell types from the previous step. The analysis is performed only on variable genes between these cell types (see below). The lowest-value cell type is removed (or values more than 0.05 below the top value), and then this step is repeated until only two cell types remain. The cell type corresponding to the top value after the last run is assigned to the single cell. Using only variable genes increases the ability to distinguish closely related cell types. Variable genes are defined as the top N genes that have a higher median expression in a cell type compared with each other cell type; we use a varying N, depending on the number of cell types used in the analysis. In the first round, N is 20 genes per cell type if there are 253 cell types (as in our ImmGen reference) or 71 genes per cell type if there are 28 cell types (as in our Blueprint + Encode reference). In the last fine-tuning step, $N = 333$. More details can be found in Supplementary Note 1 and at the SingleR Github repository[15].

Clustering in Fig. 3a was performed using Ward's hierarchical agglomerative clustering method based on the SingleR scores across all cell types. Deconvolution analysis in Fig. 2d was performed using the DeconRNA-seq package[30] using the average expression of AM samples and average expression of IM3 samples from GSE94135[29]. Differential expression analysis (Fig. 3b) was performed using the Seurat package v2.2 with default parameters.

Bulk RNA-seq. For bulk RNA-seq analysis of lung fibrosis, bleomycin (3 U kg^{-1}; Hospira) or water was instilled intratracheally to anesthetized male 129S1 mice, age 10–12 weeks. Cells gated on SiglecF$^+$CD11c$^+$ and sorted on MHCII^{hi} or MHCII^{lo} (Supplementary Fig. 6a) were collected at 2 weeks and 4 weeks after injury, or at baseline for the MHCII^{lo} compartment. RNA was collected (RNeasy Micro, Qiagen) from these samples, and total RNA quality was assessed by spectrophotometer (NanoDrop, Thermo Fisher) and the Agilent 2100 Bioanalyzer. Intact mRNA was isolated using the Dynabead mRNA Purification Kit for total RNA, according to the manufacturer's protocol (Thermo Fisher). Amplified cDNA was prepared using the NuGen Ovation RNA-Seq System V2 kit, according to the manufacturer's protocol (NuGen Technologies), and sequencing libraries were generated using the Nextera XT library preparation kit with multiplexing primers according to the manufacturer's protocol (Illumina). Library fragment size distributions were assessed using the Bioanalyzer 2100 and the DNA High-Sensitivity Chip (Agilent). Library sequence quality was assessed by sequencing single-end 50-base pair reads using the Illumina MiSeq platform, and libraries were pooled for high-throughput sequencing on the Illumina HiSeq 4000 by using equal numbers of uniquely mapped protein-coding reads. RNA sequencing was performed on HiSeq 4000 machines (Illumina), followed by de-multiplexing of raw sequencing results, trimming of adapter sequences, and alignment to the reference genome using STAR software[51]. DESeq2 was used to normalize by size factor (reads per sample) as well as by library complexity, and we then applied Wald test to determine significance of differential expression.

Public gene expression data. ImmGen raw expression data of phases 1 and 2 were downloaded as CEL files from the Gene Expression Omnibus (GEO; GSE15907 and GSE37448), processed, and normalized using the robust multi-array average procedure on probe-level data using Matlab functions. The analysis was performed using a custom CDF file obtained from Brainarray[52]. Blueprint and Encode data were processed as we described previously[53]. Raw counts from the lung reference data set were obtained from Gibbings et al.[29] (GSE94135) and Altboum et al.[28] (GSE49932). Counts were normalized and converted to TPM using R functions. The gene expression profile of GM-CSF-derived bone marrow dendritic cell subsets (available at GSE62361[19]) was analyzed using the GEO2R tool for differentially expressed genes. Top 50 upregulated by fold-change GM-CSF-cultured dendritic cell genes and top 50 GM-CSF-cultured macrophage genes were used in Fig. 1b. Single-cell RNA-seq counts from Hashimshony et al.[16] used in Fig. 1b were downloaded from the ARCHS project[54]. 10X peripheral blood mononuclear (PBMC) single-cell counts were downloaded directly from the 10X website[55].

Analysis of published human lung microarray data sets. Yang et al.[34] performed gene expression profiling for 167 lung tissues from idiopathic interstitial pneumonia subjects and 50 healthy controls. We downloaded raw CEL files from GEO (accession GSE32537) and processed them using custom CDFs from BrainArray. Normalization was performed using the robust multi-array average procedure on Affymetrix microarray data. Single-sample gene set enrichment analysis[33] was performed using human orthologs of C1 genes (Supplementary Table 3) and 15 human MHCII genes (HLA-DMA, HLA-DMB, HLA-DOA, HLA-DOB, HLA-DPA1, HLA-DPB1, HLA-DPB2, HLA-DQA1, HLA-DQA2, HLA-DQB1, HLA-DQB2, HLA-DRA, HLA-DRB1, HLA-DRB5, HLA-DRB6).

Microarray profiling in telomere dysfunction model of lung fibrosis. *SPC-creERT2/TRF1$^{flox/flox}$* mice and *TRF1$^{flox/flox}$* litter-mate controls were subjected to tamoxifen administration beginning at 8 weeks of age, as previously described[32]. Lungs perfused free of blood were harvested at 3 months and 9 months. Total RNA including small RNAs was extracted using miRNeasy Mini Kit (Qiagen). Total RNA was measured using the NanoDrop photospectrometer (NanoDrop). Agilent mouse microarray kit 8×60 K v2 (G4852B) was used. Gene expression profiling was performed by Onearray (Phalanx Bio). All of the samples passed the quality control parameters established by Agilent. Cy3-labeled antisense RNA was used for hybridization.

Single-cell dissociation for FACS. Lungs were perfused with PBS and dissociated to a single-cell solution in 10 mM HEPES RPMI containing 0.2% collagenase (Wako Pure Chemical Industries), 0.1 mg ml^{-1} Dispase II (Roche), and 2,000 U ml^{-1} DNase I (Merck). Cells were passed through a 70-µm strainer and stained at 4 °C for 30 min with the following antibodies (1:100): SiglecF-APC (catalog number 50-1702-82, clone 1RNM44N, eBioscience), MHCII-APC-Cy7 (catalog number 47-5321-82, clone M5/114.15.2, eBioscience), and CD11c-PE (catalog number 557401, clone HL3, BD Biosciences). Cells were sorted either with BD FACSAria2 or SONY SH800 FACS. Data for analytical flow were analyzed using FlowJo software.

Immunostaining and microscopy. Mouse lungs in the bleomycin model were perfused with PBS, inflated with 50% optimal cutting temperature (OCT) compound, 10% sucrose in PBS, and fixed overnight at 4 °C in 4% paraformaldehyde (PFA) before freezing in OCT. Lungs from the telomere dysfunction model were perfused with PBS and frozen in OCT. For human samples, OCT-embedded tissues were cut to 15-µm sections, and sections were fixed with 4% PFA for 15 min at room temperature. OCT-embedded tissues were cut to 15-µm sections, blocked with 3% donkey serum, 1% BSA, 0.3% TritonX, and then stained with the following antibodies (1:100 dilution): SiglecF (catalog number AF1706, R&D Systems), Mertk (catalog number AF591, R&D Systems), Pdgfrb (catalog number 14-1402-82, clone APB5, eBioscience), CD68 (catalog number ab955, clone KP1, Abcam), MafB (catalog number HPA005653, Sigma), PDGF-AA antibody (catalog number 07-1436, Millipore), and Pdgfra antibody (catalog number AF1062, R&D Systems). Secondary antibodies conjugated to Alexa fluorophores (Invitrogen and Abcam) were used at 1:200 dilution. Stained sections and two-photon images of second harmonic signal were visualized on a Zeiss LSM 780 NLO microscope. Image processing was done using ImageJ. For quantitation of Fig. 5b, regions of fibroblast accumulation were delimited by Pdgf receptor staining. For quantitation of Fig. 7a, mean fluorescent intensity of Pdgfra and Pdgfrb immunostaining was measured from cellular areas as defined by morphological tissue features. For quantitation of Fig. 7c, percentage of second harmonic signal was determined for each image, and the median across images (within patient) was defined as the threshold for categorization as fibrotic versus non-fibrotic. For hematoxylin and eosin images, staining of frozen sections was performed using standard protocols, and images of whole sections were acquired on a VERSA automated slide scanner (Leica Biosystems) equipped with an Andor Zyla 5.5 sCMOS camera (Andor Technologies) acquired under National Institutes of Health S10 grant OD021717; individual images were generated with the ImageScope software (Aperio Technologies).

Fibroblast migration assay. 3T3 mouse fibroblasts were seeded at 100% confluency into a 96-well plate with a central cell-free detection zone in the center of each well (Platypus Technologies). Then, 100 µl of conditioned media of macrophage cultures (48 h culture in RPMI containing 10% fetal bovine serum) was added per well. Pdgf-aa antibody (catalog number 07-1436, Millipore) was added at concentration 0.05 µg ml^{-1}. Cells were imaged via bright field microscopy at time points 0 h and 24 h. Image processing and migration quantification were done using ImageJ.

Fibroblast proliferation assay. Fibroblast proliferation by flow cytometric EDU assay was performed according to the manufacturer's protocol (Click-iT Flow Cytometry, Invitrogen). 3T3 mouse fibroblasts, and TdTomato^{+} macrophages sorted from *Cx3cr1-CreERT2/Rosa26-loxp-STOP-loxp-Tdtomato* mice 2 weeks after bleomycin injury, were seeded at a ratio of 1:3 in serum-free conditions in the presence of Pdgf-aa blocking antibody (catalog number 07-1436, Millipore) or IgG control (catalog number ab199376, lot GR3195151-2, Abcam). EdU was added at 48 h, and cells were suspended 2 h later for EdU detection in the TdTomato^{-} (fibroblast) fraction on a SONY SH800 flow cytometer.

Measurement of hydroxyproline. Lung hydroxyproline content was measured as previously described[6] 3 weeks after intratracheal instillation of bleomycin to sex-matched, male and female, 8–12-week-old C57BL/6J mice. Briefly, mouse lungs were incubated in 12 M HCl at 110 °C for 18 h. Aliquots of the samples reconstituted in distilled water were added to 1.4% chloramine-T in 10% isopropanol and 0.5 M sodium acetate. Erlich's solution was added, and the samples were incubated at 60 °C for 10 min. Absorbance at 562 nm was measured and adjusted according to standard curves.

Human lung tissues. Written informed consent was obtained from all subjects, and the study was approved by the University of California, San Francisco Institutional Review Board. Pulmonary fibrosis lung tissues were obtained at the time of lung transplantation from patients with a pathologic diagnosis of usual interstitial pneumonia and a consensus clinical diagnosis of idiopathic pulmonary fibrosis according to available guidelines[56]. Age-similar, non-diseased, normal lung tissues were procured from lungs not used by the Northern California Transplant Donor Network[57]. Our studies indicate that these lungs are physiologically and pathologically normal. Lung fragments were inflated with 30% OCT suspended in PBS, embedded in OCT compound, and frozen immediately after isolation in dry ice before storing at −80 °C until use.

Statistical analysis. All significance tests in this paper, unless otherwise stated, were assessed using the two-sided Wilcoxon rank-sum test.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Code availability. The SingleR open-source R package is maintained on GitHub R package and available from https://github.com/dviraran/SingleR. Code to reproduce the figures is also available in the GitHub repository.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request. RNA-seq data have been deposited in the GEO repository under accession numbers GSE111664, GSE111690, and GSE114005 and at ArrayExpress under accession number E-MTAB-7142.

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When statistical analyses are reported, confirm that the following items are present in the relevant location (e.g. figure legend, table legend, main text, or Methods section).

n/a	Confirmed	
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☐	☒	An indication of whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒	A description of all covariates tested
☐	☒	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistics including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated
☐	☒	Clearly defined error bars *State explicitly what error bars represent (e.g. SD, SE, CI)*

Our web collection on statistics for biologists may be useful.

Software and code

Policy information about availability of computer code

Data collection	No software was used to collect data.
Data analysis	Seurat v2.2 was used to process single cell data. STAR software v2.4.2a and DESeq2 v1.14.0 were used for bulk RNAseq. SingleR v1.0 was used to annotate the samples and is available at https://github.com/dviraran/SingleR. All figures and tables were generated using R, and the code for generating the figures is available at: https://github.com/dviraran/SingleR/tree/master/manuscript_figures. FlowJo v10.5.3 was used to analyze flow cytometry data. ImageJ 1.51 and ImageScope v12.3.3 were used for image analysis.

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Data

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All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

The data that support the findings of this study are available from the corresponding author upon request. RNAseq data have been deposited in the GEO depository under accession numbers GSE111664, GSE111690, and GSE114005 and at ArrayExpress under accession number E-MTAB-7142.

Field-specific reporting

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Life sciences

Study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Sample sizes were based on statistical analyses that established significant differences between groups in previous studies.
Data exclusions	Data were not excluded from the analysis.
Replication	All findings were mutiply replicated to confirm reproducibility as indicated in the text. All attempts at replication were successful.
Randomization	Groups were allocated randomly for in vitro assays and for mouse models of fibrosis. Human samples for analysis were selected randomly.
Blinding	Investigators were blinded with respect to group allocation for all assays including measurement of lung collagen and image analysis for quantitation of gap closure and cell number in photomicrographs.

Materials & experimental systems

Policy information about availability of materials

n/a	Involved in the study
☒	☐ Unique materials
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☐	☒ Research animals
☐	☒ Human research participants

Antibodies

Antibodies used	SiglecF- APC (eBioscience, cat#50-1702-82, clone 1RNM44N, lot 4285690, dilution 1:100), MHCII-APC-Cy7 (eBioscience, cat#47-5321-82, clone M5/114.15.2, lot 4310394, dilution 1:100), CD11c-PE (BD Biosciences, cat#557401, clone HL3, lot 7012770, dilution 1:100), SiglecF (R&D Systems, cat#AF1706, lot KPA011707A, dilution 1:100), Mertk (R&D Systems, cat#AF591, lot DGS05, dilution 1:100), PDGFRbeta (eBioscience, cat#14-1402-82, clone APB5, lot 4322162, dilution 1:100), CD68 (Abcam, clone KP1, cat#ab955, lot GR3192728, dilution 1:100), MafB (Sigma, cat#HPA005653, lot 6117647, dilution 1:100), Alexa fluorophore secondary antibodies (Invitrogen and Abcam), PDGF-AA (Millipore, 07-1436, lot 3100458, dilution 1:100 or 0.05 microgram/mL), Rabbit IgG isotype control (Abcam, cat#ab199376, lot GR3195151-2, 0.05microgram/mL), PDGFRalpha (R&D Systems, cat#AF1062, lot HMQ0217091, dilution 1:100). Secondary antibodies, all Donkey used at dilution 1:200: anti-goat 555 (Invitrogen, cat#A21432, lot 1249013), anti-goat 647 (Invitrogen, cat#A21447, lot 1977345), anti-rabbit 555 (Abcam, cat#ab150062, lot GR297073-1), anti-rabbit 647 (Abcam, cat#ab150063, lot GR-289647-4), anti-rat 555 (Abcam, cat#ab150154, lot GR2901144-1), anti-rat 647 (Abcam, cat#ab150155, lot GR303632-3).
Validation	All antibodies used were commercially available with accompanying validation information available at vendor websites.

Eukaryotic cell lines

Policy information about cell lines

Cell line source(s)	FIbroblasts 3T3 cells from ATCC.
Authentication	The cell line was used directly from the vendor. No further authentication was performed.
Mycoplasma contamination	Mycoplasma testing was not performed after purchase.
Commonly misidentified lines (See ICLAC register)	We did not use any commonly misidentified cell lines per the current ICLAC register.

Research animals

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research

Animals/animal-derived materials	Mouse, 129S1 and C57BL/6 ages 8-12 weeks of both sexes were used.

Human research participants

Policy information about studies involving human research participants

Population characteristics	Lung samples were acquired from patients with idiopathic pulmonary fibrosis (IPF) at the time of lung transplant or from control lungs when not used by the Northern California Donor Transplant network. There was a total of 6 samples from patients with IPF ranging in age from 49 to 67, 5 females and 1 male. At the time of transplant, one patient was on treatment with pirfenidone and 1 with nintedanib. For unused donor controls, demographic data were not collected.

Method-specific reporting

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ Magnetic resonance imaging

Flow Cytometry

Plots

Confirm that:

☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).

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☒ All plots are contour plots with outliers or pseudocolor plots.

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Methodology

Sample preparation	Mouse lungs were collagenase digested into single cell suspensions as described in Methods.
Instrument	BD FACSAria2 or SONY SH800 FACS
Software	FlowJo
Cell population abundance	Cell populations depicted in Supplementary Figure 6.
Gating strategy	Gating strategy depicted in Supplementary Figure 6.

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